

Final Project: Machine Learning & Deep Learning

In-depth Exploration of a Real-World Data Science Problem

K. Kadri

Outline

- 1 General Context
- 2 Learning Objectives
- 3 Tuning & Explainability
- 4 Deliverables
- 5 Evaluation Criteria

Project Objective

- Conduct a full Machine Learning project (with a bit of Deep Learning).
- Work on a real dataset, freely chosen by the student.
- Explore an original angle or an in-depth problem.

Important

The goal is not only to obtain high accuracy, but to conduct a **complete Data Science workflow**.

Targeted Competencies

- Preprocessing, Feature Engineering, Feature Selection.
- Training multiple ML models.
- Hyperparameter tuning.
- Explainability (SHAP, feature importances).
- Building a simple DL model (ANN or CNN).
- Critical analysis, limitations, and conclusion.

- Identify structure, distribution, outliers.
- Clearly define the problem: classification, regression, clustering.

Key Question

What is the true nature of the problem you are trying to solve?

Complete ML Pipeline

- Feature Engineering
- Standardization / Normalization
- Categorical Encoding
- Feature Selection
- Cross-validation

Good Practice

Use a clear, reproducible, well-justified pipeline.

Required ML Models

- Regression (linear or logistic, depending on the dataset)
- KNN
- Decision Tree
- Random Forest
- SVM
- Structured comparison of all models

Deep Learning (minimal model)

- ANN (for tabular data) **or**
- CNN (for images)
- Analysis of loss curves: overfitting, regularization

Advice

Keep the model simple but well explained.

Hyperparameter Tuning

- GridSearch
- RandomSearch
- (Optional) Optuna / Bayesian Optimization

Expected

Compare **before/after** tuning for at least one model.

- SHAP Summary Plot
- SHAP Local (Force Plot)
- Feature Importances (RF or Gradient Boosting)

Key Skill

Know how to interpret results, not only compute them.

Structure of the Final Report

- 1 Problem statement
- 2 Dataset description
- 3 Exploratory Data Analysis (EDA)
- 4 Feature Engineering
- 5 ML models evaluated
- 6 Hyperparameter tuning
- 7 Explainability (SHAP)
- 8 DL model
- 9 Conclusion + limitations

Essential

The project must be clear, reproducible, structured, and well argued.

- Clean code (Notebook + Python scripts).
- Rigorous experimentation.
- Relevant visualizations.
- Thoughtful interpretation.
- Critical approach to limitations.

A — Technical Rigor (40%)

- Clean, reproducible ML pipeline.
- Correct use of algorithms.
- Coherent preprocessing.
- Functional DL model (even simple).

B — Feature Engineering (20%)

- Smart creation of new features.
- Coherent selection (statistical or model-based).
- Impact on performance clearly justified.

Note

A project with few but well-engineered features is better than a complex model poorly explained.

- Correct use of SHAP.
- Interpretation of key variables.
- Error analysis.

D — In-Depth Exploration (20%)

- Creative approach to the problem.
- Deep analysis: biases, advanced metrics, thresholds, residuals.
- Rigorous cross-validation.
- Personal initiative beyond automation.

Premium Criterion

What matters is not raw accuracy, but your **depth of analysis and understanding**.